# Interpretable Machine Learning

# Feature Importance
# Intro to Loss-based Feature Importance

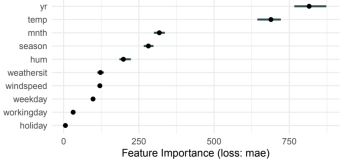
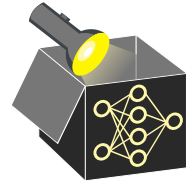



**Figure:** Bike Sharing Dataset

**Learning goals**

- Understand motivation for feature importance
- Develop an intuition for possible use-cases
- Know characteristics of feature importance methods

# MOTIVATION



- **Feature effects** describe how a feature $x_j$ influences the prediction $\hat{y}$
  - requires one plot per feature (e.g., PDPs, ALEs)
  - purely model-based; ignores true target $y$

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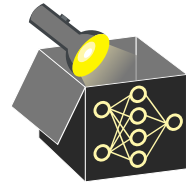

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  - requires one plot per feature (e.g., PDPs, ALEs)
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  - condenses information into one number per feature
  - typically compares prediction errors (involves $y$) with/without feature

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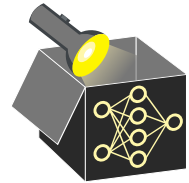

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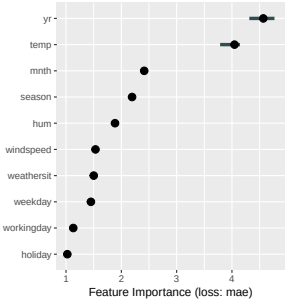
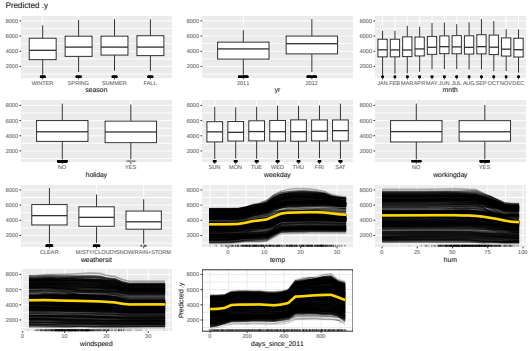
  - condenses information into one number per feature
  - typically compares prediction errors (involves $y$) with/without feature
- **Clarification:** By *feature importance*, we mean *loss-based* methods that assess a feature's impact via changes in *prediction error*.
  $\rightsquigarrow$ Other notions exist (e.g., variance-based methods; see ▸ "Greenwell et al." 2020).

# EXAMPLE

Feature importance provides a condensed summary of feature relevance w.r.t. performance

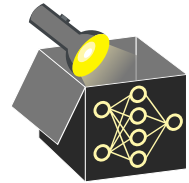
- Fit random forest on bike sharing data
- Left: Feature importance ranking by permutation feature importance (PFI)
- Right: Feature effects for all features



# FEATURE IMPORTANCE - DIFFERENCES

▸ **"Ewald et al." 2024**



Many loss-based feature importance methods exist, which mainly differ in

**(1) How they "remove" or "perturb" the feature of interest (FOI)** $X_j$

- **Remove** $X_j$ **and refit:** Drop the $X_j$ and retrain model without it
- **Perturb** $X_j$**:** Replace $X_j$ by $\tilde{X}_j$ sampled from *marginal*/*conditional* distrib.
- **Marginalize** $X_j$**:** integrate out $X_j$ via *marginal* or *conditional* distribution

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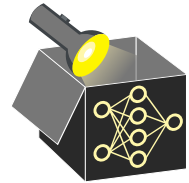

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**(2) How they compare model performance before and after feat. removal**

- **Compare "reduced model" without FOI vs. full model**:
  Measure drop in performance when FOI is "removed"
  $\rightsquigarrow$ Similar idea as backward feature elimination
- **Compare "empty model" (no features) vs. model with only FOI**:
  Measure gain in performance when only FOI is used
  $\rightsquigarrow$ Similar idea as forward feature selection
- **Compare models with/without FOI across different feature sets**:
  Measure average contrib. when FOI joins any feat. set (Shapley-based)

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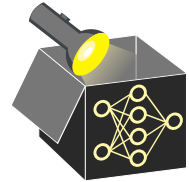

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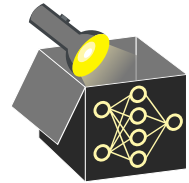
*Depending on the different removal/perturbation and comparison strategies, feat. imp. methods provide insight into different aspects of model and data.*

# POTENTIAL INTERPRETATION GOALS



Feature importance methods provide condensed insights, but only into specific aspects of model and data. Interpretation goals often differ and typically address non-overlapping questions (except for special cases).

For example, one may be interested in getting insight into whether the ...

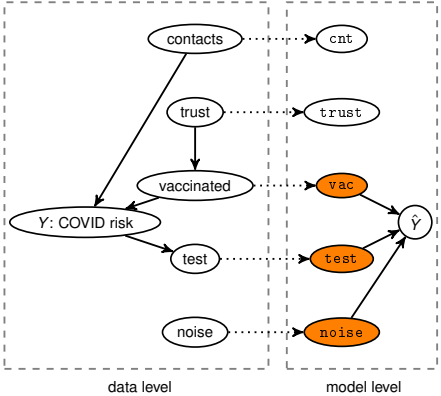
**(1)** feature $x_j$ is causal for the prediction?

**(2)** feature $x_j$ contains prediction-relevant information about $y$?

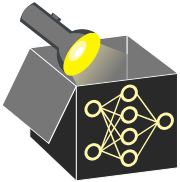
**(3)** model requires access to $x_j$ to achieve its prediction performance?

# EXAMPLE: CAUSAL FOR THE PREDICTION (1)

A feature may be causal for $\hat{y}$ (1) without containing prediction-relevant information about $y$ (2)




*Example:* overfitting due to noisy features



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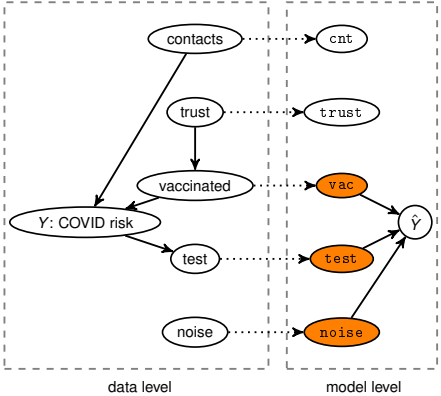
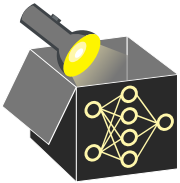



*Example:* overfitting due to noisy features

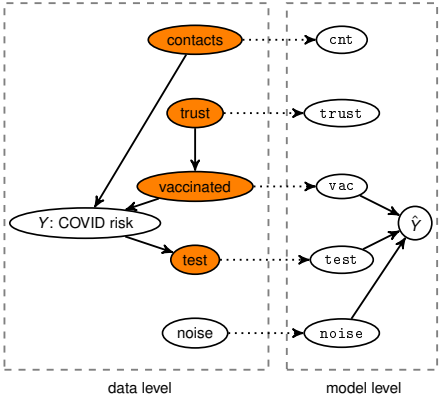
- All features used by the model are of interest
- Here: Model uses feature `noise`, although it does not contain prediction-relevant information about $y$ (data level)

$\Rightarrow$ Overfitted models may use many noise features which are deemed relevant on model level (but not on data level)

# EXAMPLE: PREDICTION-RELEVANT INFORMATION (2)



A feature may contain prediction-relevant information (2) without causing the prediction (1)

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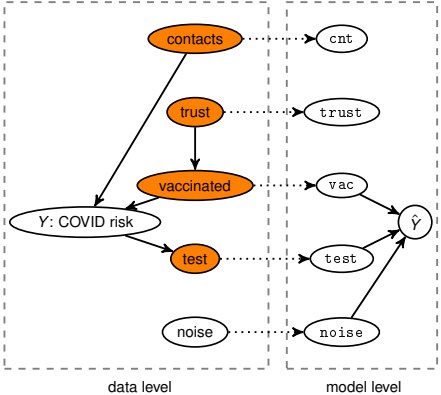


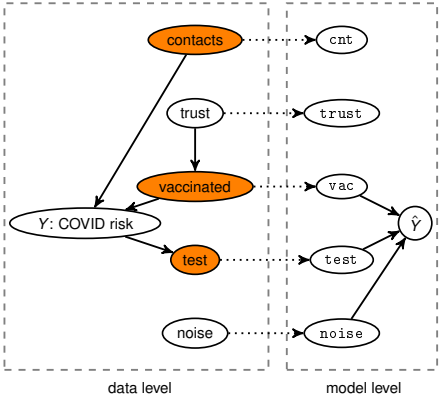

*Example:* underfitting, model multiplicity

- All prediction-relevant features for $y$ are of interest
- Example: All features that are directly or indirectly (i.e., via another feature) connected to $y$

$\Rightarrow$ Underfitted models may ignore prediction-relevant features such as `contacts` here

# EXAMPLE: REQUIRES ACCESS TO FEATURE (3)

A feature may contain prediction-relevant information (2), without the model requiring access to the feature for (optimal) prediction performance (3)

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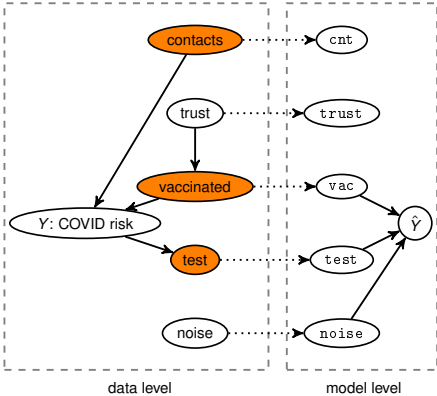
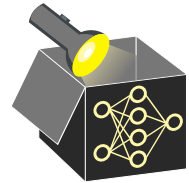



*Example:* correlations, confounding

- All unique prediction-relevant features for $y$ are of interest
- Example: All features that are directly connected to $y$

$\Rightarrow$ `trust` and `vaccinated` may be correlated but only `vaccinated` is directly connected to $y$



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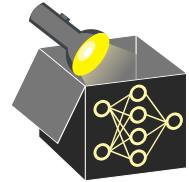

For example, one may be interested in getting insight into whether the ...

**(1)** feature $x_j$ is causal for the prediction?

- A symptom may help predict a disease ($\rightsquigarrow$ causal for $\hat{y}$)
- Intervening on symptom may not affect disease ($\rightsquigarrow$ not causal for $y$)

**(2)** feature $x_j$ contains prediction-relevant information about $y$?

- $x_j$ helps predict $y$ (e.g., conditional expectation) w.r.t. performance
- If $x_j \perp\!\!\!\perp y$, then $\mathbb{E}[y|x_j] = \mathbb{E}[y]$ and $x_j$ and $y$ have 0 mutual info.
  $\rightsquigarrow$ $x_j$ carries no prediction-relevant information

**(3)** model requires access to $x_j$ to achieve its prediction performance?

- $x_j$ helps predict $y$ w.r.t. performance, compared to using only $x_{-j}$
- If $x_j \perp\!\!\!\perp y|x_{-j}$, then $\mathbb{E}[y|x_{-j}] = \mathbb{E}[y|x_j, x_{-j}]$
  $\rightsquigarrow$ $x_j$ does not contribute unique prediction-relevant information about $y$
- **Note:** A model may rely on features that can be replaced with others, e.g., if $\mathbb{E}[y \mid x_1] \neq \mathbb{E}[y]$ and $\mathbb{E}[y \mid x_1] = \mathbb{E}[y \mid x_1, x_2]$, a random forest may ignore $x_1$ in splitting and rely on $x_2$ instead (despite $x_1$ being informative).